Glasser's Master Theorem

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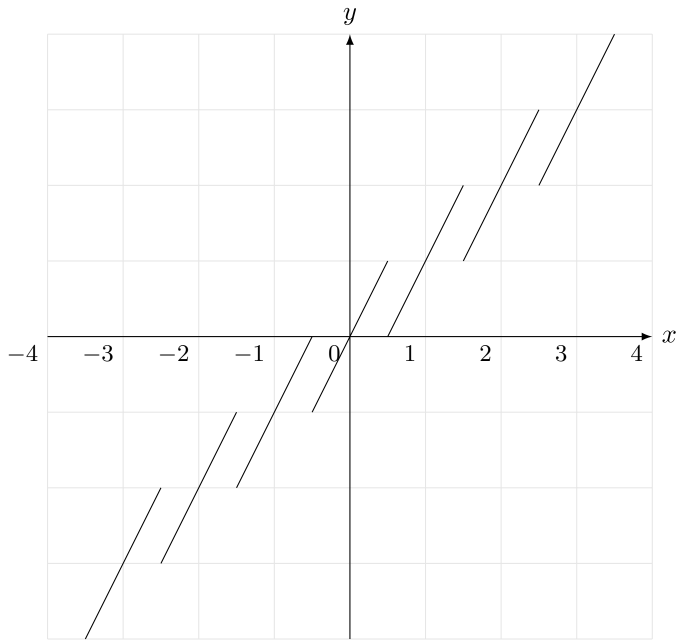
Definition -1.1. A function $f : \mathbb{R} \to \mathbb{R}$ **preserves the Lebesgue measure on** $\mathbb{R}$ if for every Lebesgue measurable $A \subseteq \mathbb{R}$, we have $\mu(f^{-1}(A)) = \mu(A)$, where $f^{-1}(A) = \{x \in \mathbb{R} \mid f(x) \in A\}$ is the pre-image of A under f.

Examples.

1. For any $a \in \mathbb{R}$, the function $f(x) = x + a$ preserves the Lebesgue measure on $\mathbb{R}$.
2. The function $f(x) = 2x + \lfloor x \rfloor - \lfloor 2x \rfloor$ preserves the Lebesgue measure on $\mathbb{R}$. This is because for each $n \in \mathbb{Z}$, the interval $[n, n+1)$ is mapped to two copies of itself, each with a slope of 2 (so the pre-image of any subinterval of $[n, n+1)$ is mapped to two subintervals, each with half the length). Likewise, for any $m \in \mathbb{Z}\backslash\{0\}$, the function $f(x) = mx + \lfloor x \rfloor - \lfloor mx \rfloor$ preserves the Lebesgue measure on $\mathbb{R}$.



0 The Cauchy-Schlömilch Transformation

Lemma 0.1 (Cauchy-Schlömilch Transformation)**.** Suppose $c > 0$ and define $f : \mathbb{R}\backslash\{0\} \to \mathbb{R}$ as follows:

$$f(x) = x - \frac{c}{x}$$

Then f preserves the Lebesgue measure on $\mathbb{R}$.

Proof. Suppose $t \in \mathbb{R}$. We first solve the inequality $f(x) < t$, i.e. $x - \frac{c}{x} < t$.

- If $x > 0$, this implies $x^2 - tx - c < 0$, and so $\frac{t-\sqrt{t^2+4c}}{2} < x < \frac{t+\sqrt{t^2+4c}}{2}$. Since $\frac{t-\sqrt{t^2+4c}}{2} < 0$, this simplifies to $0 < x < \frac{t+\sqrt{t^2+4c}}{2}$.
- If $x < 0$, this implies $x^2 - tx - c > 0$, and so $x < \frac{t-\sqrt{t^2+4c}}{2}$ or $x > \frac{t+\sqrt{t^2+4c}}{2}$. Since $\frac{t+\sqrt{t^2+4c}}{2} > 0$, this simplifies to $x < \frac{t-\sqrt{t^2+4c}}{2}$.

All in all, we have:

$$f(x) < t \iff x < \frac{t-\sqrt{t^2+4c}}{2} \text{ or } 0 < x < \frac{t+\sqrt{t^2+4c}}{2}$$

Similarly:

$$f(x) > t \iff \frac{t-\sqrt{t^2+4c}}{2} < x < 0 \text{ or } x > \frac{t+\sqrt{t^2+4c}}{2}$$

Note that for all $t \in \mathbb{R}$, we have:

$$\frac{\partial}{\partial t}\left(\frac{t \pm \sqrt{t^2+4c}}{2}\right) = \frac{1}{2} \pm \frac{t}{2\sqrt{t^2+4c}} \geq \frac{1}{2} - \frac{|t|}{2\sqrt{t^2+4c}} > \frac{1}{2} - \frac{1}{2} = 0$$

Thus the functions $t \mapsto \frac{t\pm\sqrt{t^2+4c}}{2}$ are both strictly increasing in t.
Now suppose $\alpha, \beta \in \mathbb{R}$, $\alpha < \beta$. Then we have:

$$\alpha < f(x) < \beta \iff \left(x < \frac{\beta - \sqrt{\beta^2+4c}}{2} \text{ or } 0 < x < \frac{\beta + \sqrt{\beta^2+4c}}{2}\right)$$
$$\text{and } \left(\frac{\alpha - \sqrt{\alpha^2+4c}}{2} < x < 0 \text{ or } x > \frac{\alpha + \sqrt{\alpha^2+4c}}{2}\right)$$

This reduces to:

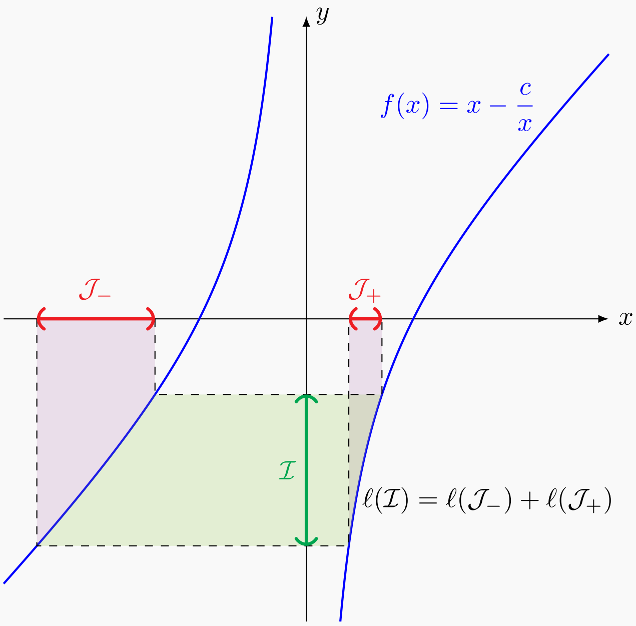
$$\frac{\alpha - \sqrt{\alpha^2+4c}}{2} < x < \frac{\beta - \sqrt{\beta^2+4c}}{2} \text{ or } \frac{\alpha + \sqrt{\alpha^2+4c}}{2} < x < \frac{\beta + \sqrt{\beta^2+4c}}{2}$$

The union of these intervals has length:

$$\frac{\beta - \sqrt{\beta^2+4c}}{2} - \frac{\alpha - \sqrt{\alpha^2+4c}}{2} + \frac{\beta + \sqrt{\beta^2+4c}}{2} - \frac{\alpha + \sqrt{\alpha^2+4c}}{2} = \frac{\beta}{2} - \frac{\alpha}{2} + \frac{\beta}{2} - \frac{\alpha}{2} = \beta - \alpha$$

Thus $\mu(f^{-1}((\alpha,\beta))) = \mu((\alpha,\beta))$ for all $\alpha, \beta \in \mathbb{R}$, $\alpha < \beta$. By the Carathéodory extension theorem, we have $\mu(f^{-1}(A)) = \mu(A)$ for every Lebesgue measurable $A \subseteq \mathbb{R}$. Thus f preserves the Lebesgue measure on $\mathbb{R}$. 😎

Cauchy-Schlömilch Transformation



For any interval on the y-axis, the length of the interval is equal to the sum of the lengths of the corresponding intervals on the x-axis.

1 Glasser's Master Theorem

The following theorem was first stated in 1983 by M. Lawrence Glasser (1933–).

Theorem 1.1 (Glasser's Master Theorem)**.** Suppose $a_1, a_2, \cdots, a_n \in \mathbb{R}$, $c_1, c_2, \ldots, c_n > 0$ and $b \in \mathbb{R}$. Define $f : \mathbb{R}\backslash\{a_1, a_2, \ldots, a_n\}$ as follows:

$$f(x) = x - b - \frac{c_1}{x-a_1} - \frac{c_2}{x-a_2} - \cdots - \frac{c_n}{x-a_n}$$

Then f preserves the Lebesgue measure on $\mathbb{R}$.

Proof. Assume without loss of generality that $a_1 < a_2 < \cdots < a_n$. Define $a_0 = -\infty$ and $a_{n+1} = \infty$, and for each $k \in \{0, 1, \ldots, n\}$, define $I_k = (a_k, a_{k+1})$. Note that for all $x \in \mathbb{R}\backslash\{a_1, a_2, \ldots, a_n\}$, we have:

$$f'(x) = 1 + \frac{c_1}{(x-a_1)^2} + \frac{c_2}{(x-a_2)^2} + \cdots + \frac{c_n}{(x-a_n)^2} > 1$$

We also have $\lim_{x \to a_k^-} f(x) = \infty$ and $\lim_{x \to a_k^+} = -\infty$ for all $k \in \{0, 1, \ldots, n\}$. Thus for each $k \in \{0, 1, \ldots, n\}$, the restriction of f to I_k is a bijection from I_k to $\mathbb{R}$.

Define $g_k : \mathbb{R} \to I_k$ as the inverse of the restriction of f to I_k. Then for each $y \in \mathbb{R}$, the equation $f(x) = y$ has exactly $n+1$ solutions, namely $g_0(y), g_1(y), \ldots, g_n(y)$. Multiplying both sides of $f(x) - y = 0$ by $(x - a_1)(x - a_2) \cdots (x - a_n)$, we get:

$$(x - a_1)(x - a_2) \cdots (x - a_n)\left(x - b - \frac{c_1}{x - a_1} - \frac{c_2}{x - a_2} - \cdots - \frac{c_n}{x - a_n}\right) = 0$$
$$(x - a_1)(x - a_2) \cdots (x - a_n)(x - b - y) + P_{n-1}(x) = 0$$

Where $P_{n-1}(x)$ is a polynomial in x of degree at most $n - 1$. Since the left side is a monic polynomial of degree $n+1$ that vanishes at $g_0(y), g_1(y), \ldots, g_n(y)$, it must be equal to $(x - g_0(y))(x - g_1(y)) \cdots (x - g_n(y))$. Thus we have:

$$(x - a_1)(x - a_2) \cdots (x - a_n)(x - b - y) + P_{n-1}(x) = (x - g_0(y))(x - g_1(y)) \cdots (x - g_n(y))$$

Comparing the coefficients of x^n on each side, we get:

$$-(a_1 + a_2 + \cdots + a_n + b + y) = -(g_0(y) + g_1(y) + \cdots + g_n(y))$$

Since each g_k is strictly increasing, for any $y, z \in \mathbb{R}$, $y < z$, we have $f(x) \in (y, z)$ if and only if $x \in (g_k(y), g_k(z))$ for some $k \in \{0, 1, \ldots, n\}$. This yields:

$$\begin{aligned}
\mu(f^{-1}((y, z))) &= \mu\left(\bigsqcup_{k=0}^{n} (g_k(y), g_k(z))\right) \\
&= \sum_{k=0}^{n} \mu((g_k(y), g_k(z))) \\
&= \sum_{k=0}^{n} (g_k(z) - g_k(y)) \\
&= (a_1 + a_2 + \cdots + a_n + b + z) - (a_1 + a_2 + \cdots + a_n + b + y) \\
&= z - y \\
&= \mu((y, z))
\end{aligned}$$

Thus $\mu(f^{-1}((y, z))) = \mu((y, z))$ for all $y, z \in \mathbb{R}$, $y < z$. By the Carathéodory extension theorem, we have $\mu(f^{-1}(A)) = \mu(A)$ for every Lebesgue measurable $A \subseteq \mathbb{R}$. Thus f preserves the Lebesgue measure on $\mathbb{R}$. 😎

Remark. If $n = 1$, i.e. $f(x) = x - b - \frac{c}{x-a}$ (where $c > 0$), we can explicitly compute the inverse of f:

$$f^{-1}(y) = \frac{y + b + a \pm \sqrt{(y + b - a)^2 + 4c}}{2}$$

The $\pm$ corresponds to the fact that there are two values of x for each value of y, one with $x > b$ and one with $x < b$.

Theorem 1.2. Suppose $f : \mathbb{R} \to \mathbb{R}$ preserves the Lebesgue measure on $\mathbb{R}$. Then for any $\phi \in L^1(\mathbb{R})$, we have:

$$\int_{\mathbb{R}} \phi(f(x))\, d\mu(x) = \int_{\mathbb{R}} \phi(x)\, d\mu(x)$$

Proof. If $\phi = 1_A$ is an indicator function of some Lebesgue measurable $A \subseteq \mathbb{R}$, this reduces to $\mu(f^{-1}(A)) = \mu(A)$, which is true as f preserves the Lebesgue measure on $\mathbb{R}$ by assumption. By linearity, the result holds for simple functions (linear combinations of indicator functions), and by approximation, it holds for all integrable functions.

$$\int_{\mathbb{R}} \phi(f(x))\, d\mu(x)$$

😎

In the case of $f(x) = x - \frac{c}{x}$ (the Cauchy-Schlömilch transformation), we can give a much simpler proof:

Proof. Substituting $u = -\frac{c}{x}$, we get:

$$\int_{-\infty}^{\infty} \phi\left(x - \frac{c}{x}\right) dx = c \int_{-\infty}^{\infty} \frac{1}{u^2} \phi\left(u - \frac{c}{u}\right) du$$

We now add the left side of the above equation to both sides:

$$\begin{aligned}
2 \int_{-\infty}^{\infty} \phi\left(x - \frac{c}{x}\right) dx &= \int_{-\infty}^{\infty} \phi\left(x - \frac{c}{x}\right) dx + c \int_{-\infty}^{\infty} \frac{1}{x^2} \phi\left(x - \frac{c}{x}\right) dx \\
&= \int_{-\infty}^{\infty} \left(1 + \frac{c}{x^2}\right) \phi\left(x - \frac{c}{x}\right) dx
\end{aligned}$$

We now substitute $y = x - \frac{c}{x}$. Note that as x runs from $-\infty$ to ∞, y runs from $-\infty$ to ∞ *twice* (once when $x < 0$ and once when $x > 0$). This introduces a factor of 2 below:

$$2\int_{-\infty}^{\infty} \phi\left(x - \frac{c}{x}\right) dx = 2\int_{-\infty}^{\infty} \phi(y)\, dy$$

Dividing by 2 yields the desired result. 😎

Example.

$$\int_{-\infty}^{\infty} \frac{x^2}{x^4+1}\, dx = \int_{-\infty}^{\infty} \frac{1}{\left(x-\frac{1}{x}\right)^2+2}\, dx = \int_{-\infty}^{\infty} \frac{1}{x^2+2}\, dx = \frac{\pi}{\sqrt{2}}$$

Example. Unfortunately, $\frac{1}{x^4+1}$ does not have a simple expression in terms of $x - \frac{1}{x}$. However, note that:

$$\frac{x^2+1}{x^4+1} = \frac{1+\frac{1}{x^2}}{\left(x-\frac{1}{x}\right)^2+2}$$

This allows us to easily find $\int_{-\infty}^{\infty} \frac{x^2+1}{x^4+1}\, dx$. Note that we are *not* using Glasser's master theorem here, we are simply substituting $u = x - \frac{1}{x}$.

$$\int_{-\infty}^{\infty} \frac{x^2+1}{x^4+1}\, dx = \int_{-\infty}^{\infty} \frac{1+\frac{1}{x^2}}{\left(x-\frac{1}{x}\right)^2+2}\, dx = 2\int_{-\infty}^{\infty} \frac{1}{u^2+2}\, du = 2\left(\frac{\pi}{\sqrt{2}}\right) = \sqrt{2}\pi$$

The factor of 2 comes from the fact that as x runs from $-\infty$ to ∞, u runs from $-\infty$ to ∞ twice. Thus, we have:

$$\int_{-\infty}^{\infty} \frac{1}{x^4+1}\, dx = \int_{-\infty}^{\infty} \frac{x^2+1}{x^4+1}\, dx - \int_{-\infty}^{\infty} \frac{x^2}{x^4+1}\, dx = \sqrt{2}\pi - \frac{\pi}{\sqrt{2}} = \frac{\pi}{\sqrt{2}}$$

Example.

$$\int_{-\infty}^{\infty} \frac{x^4}{x^8+1}\, dx = \int_{-\infty}^{\infty} \frac{1}{\left(x-\frac{1}{x}\right)^4+4\left(x-\frac{1}{x}\right)^2+2}\, dx = \int_{-\infty}^{\infty} \frac{1}{x^4+4x^2+2}\, dx = \frac{\sqrt{4-2\sqrt{2}}}{4}\pi$$

$$\int_{-\infty}^{\infty} \frac{x^4}{x^8+x^4+1}\, dx = \int_{-\infty}^{\infty} \frac{1}{\left(x-\frac{1}{x}\right)^4+4\left(x-\frac{1}{x}\right)^2+3}\, dx = \int_{-\infty}^{\infty} \frac{1}{x^4+4x^2+3}\, dx = \frac{3-\sqrt{3}}{6}\pi$$

$$\int_{-\infty}^{\infty} \frac{x^4}{x^8+ax^4+1}\, dx = \int_{-\infty}^{\infty} \frac{1}{\left(x-\frac{1}{x}\right)^4+4\left(x-\frac{1}{x}\right)^2+a+2}\, dx = \int_{-\infty}^{\infty} \frac{1}{x^4+4x^2+a+2}\, dx = \sqrt{\frac{\sqrt{a+2}-2}{2(a^2-4)}}\pi$$

$$\int_{-\infty}^{\infty} \frac{x^6+x^4}{x^{10}+1}\, dx = \int_{-\infty}^{\infty} \frac{1}{\left(x-\frac{1}{x}\right)^4+3\left(x-\frac{1}{x}\right)^2+1}\, dx = \int_{-\infty}^{\infty} \frac{1}{x^4+3x^2+1}\, dx = \frac{\pi}{\sqrt{5}}$$

$$\int_{-\infty}^{\infty} \frac{x^4}{(x^4+1)(x^4-x^2+1)}\, dx = \int_{-\infty}^{\infty} \frac{1}{\left(x-\frac{1}{x}\right)^4+3\left(x-\frac{1}{x}\right)^2+2}\, dx = \int_{-\infty}^{\infty} \frac{1}{x^4+3x^2+2}\, dx = \left(1-\frac{1}{\sqrt{2}}\right)\pi$$

Example.

$$\int_{-\infty}^{\infty} e^{-\left(x-\frac{1}{x}\right)^2}\, dx = \int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi}$$

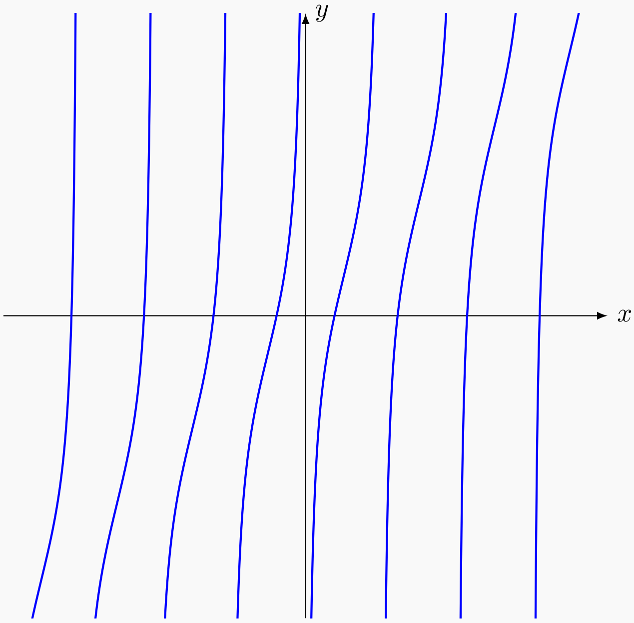
Example.

$$\int_{-\infty}^{\infty} \operatorname{sech}\left(x-\frac{1}{x}\right) dx = \int_{-\infty}^{\infty} \operatorname{sech}(x)\, dx = \pi$$

2 Generalization to infinitely many poles

We can generalize Glasser's master theorem to more exotic functions.

The function $f(x) = x - \cot(x)$



The function $f : \mathbb{R}\backslash\pi\mathbb{Z} \to \mathbb{R}$, $f(x) = x - \cot(x)$ is qualitatively very similar to the functions in Glasser's master theorem (except that this function has *infinitely* many poles).

It turns out that f also preserves the Lebesgue measure on $\mathbb{R}$. This is because:

$$\cot(x) = \cdots + \frac{1}{x-3\pi} + \frac{1}{x-2\pi} + \frac{1}{x-\pi} + \frac{1}{x} + \frac{1}{x+\pi} + \frac{1}{x+2\pi} + \frac{1}{x+3\pi} + \cdots$$

More formally, for all $x \in \mathbb{R}\backslash\pi\mathbb{Z}$, we have:

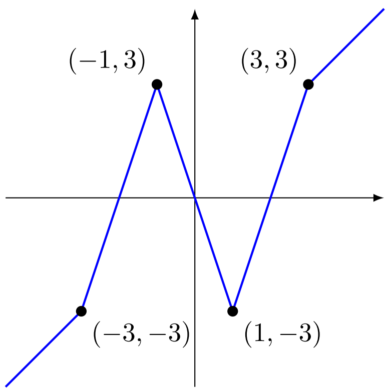
$$\cot(x) = \lim_{n\to\infty} \sum_{k=-n}^{n} \frac{1}{x + k\pi}$$

https://math.stackexchange.com/a/1015739

There are many more functions that preserve the Lebesgue measure on $\mathbb{R}$. The example below is continuous and piecewise linear.

Example. Define $f : \mathbb{R} \to \mathbb{R}$ as follows:

$$f(x) = \begin{cases} x & x < -3 \text{ or } x > 3 \\ 3x+6 & -3 < x < -1 \\ -3x & -1 \leq x \leq 1 \\ 3x-6 & 1 < x < 3 \end{cases}$$



Then f is continuous, piecewise linear and preserves the Lebesgue measure on $\mathbb{R}$. Informally, this is because $f(x) = x$ except on $(-3, 3)$, where it traverses the interval $(-3, 3)$ three times, each with a slope of ± 3. More examples can be constructed similarly, e.g. by traversing an interval 5 times, each with a slope of ± 5.